CS 663 : Digital Image Processing

Assignment 3

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Q1 – Bison: Segmentation

Mean-shift segmentation (hr=40, hs=250)

Input image



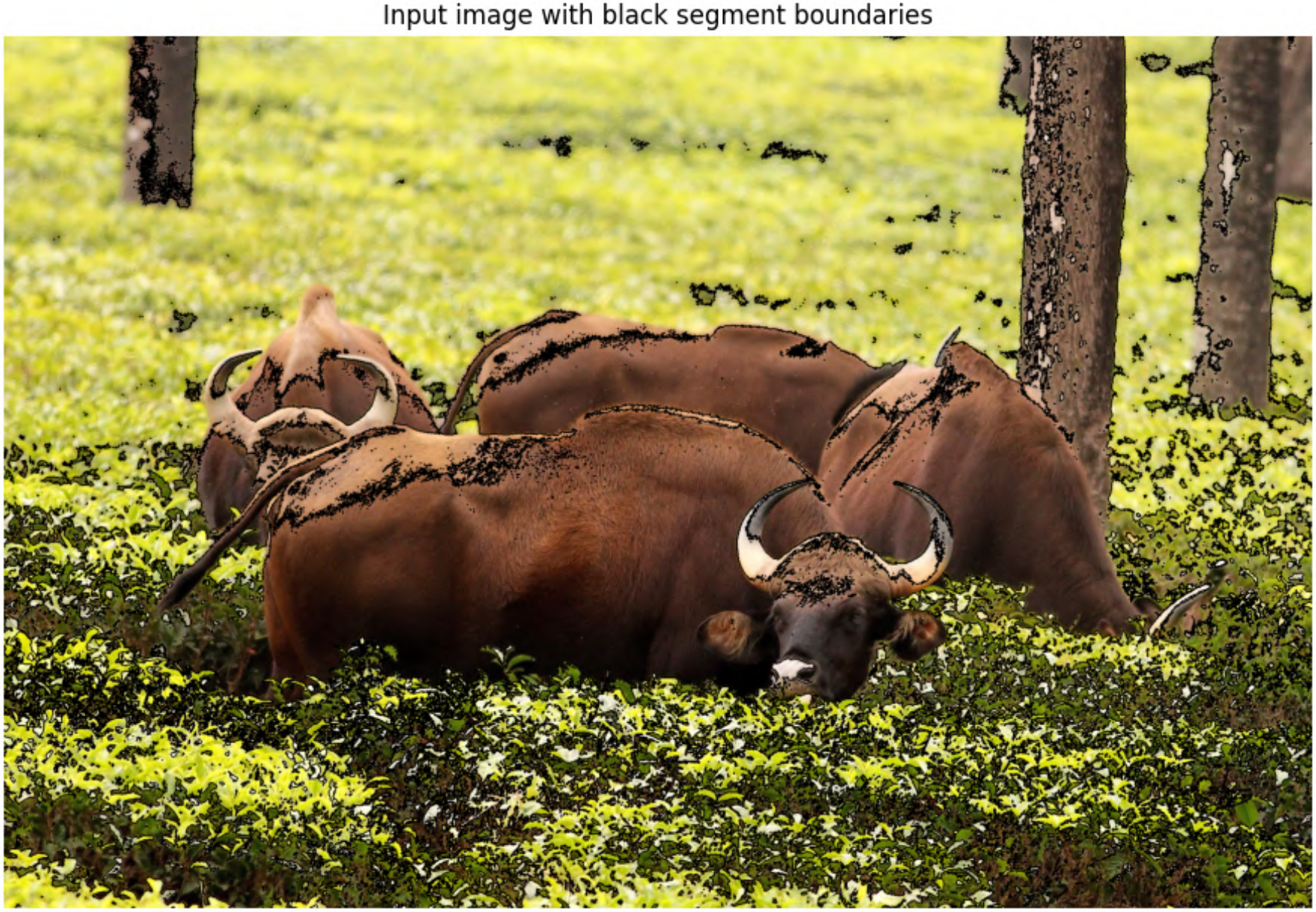
Mean-shift segmentation (5 segments)



Q1 – Bison: Label Map



Q1 – Bison: Boundary Overlay



Q1 – Ghats: Segmentation

Mean-shift segmentation (hr=25, hs=300)

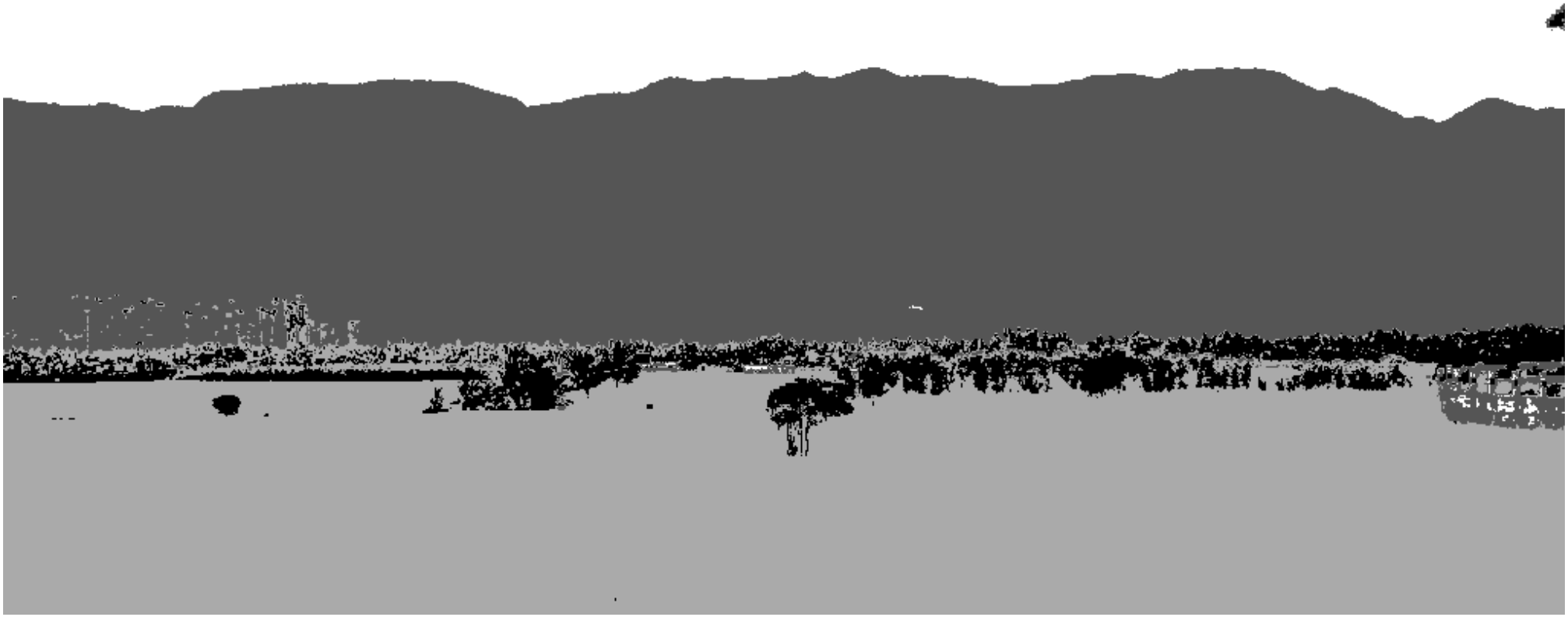
Input image



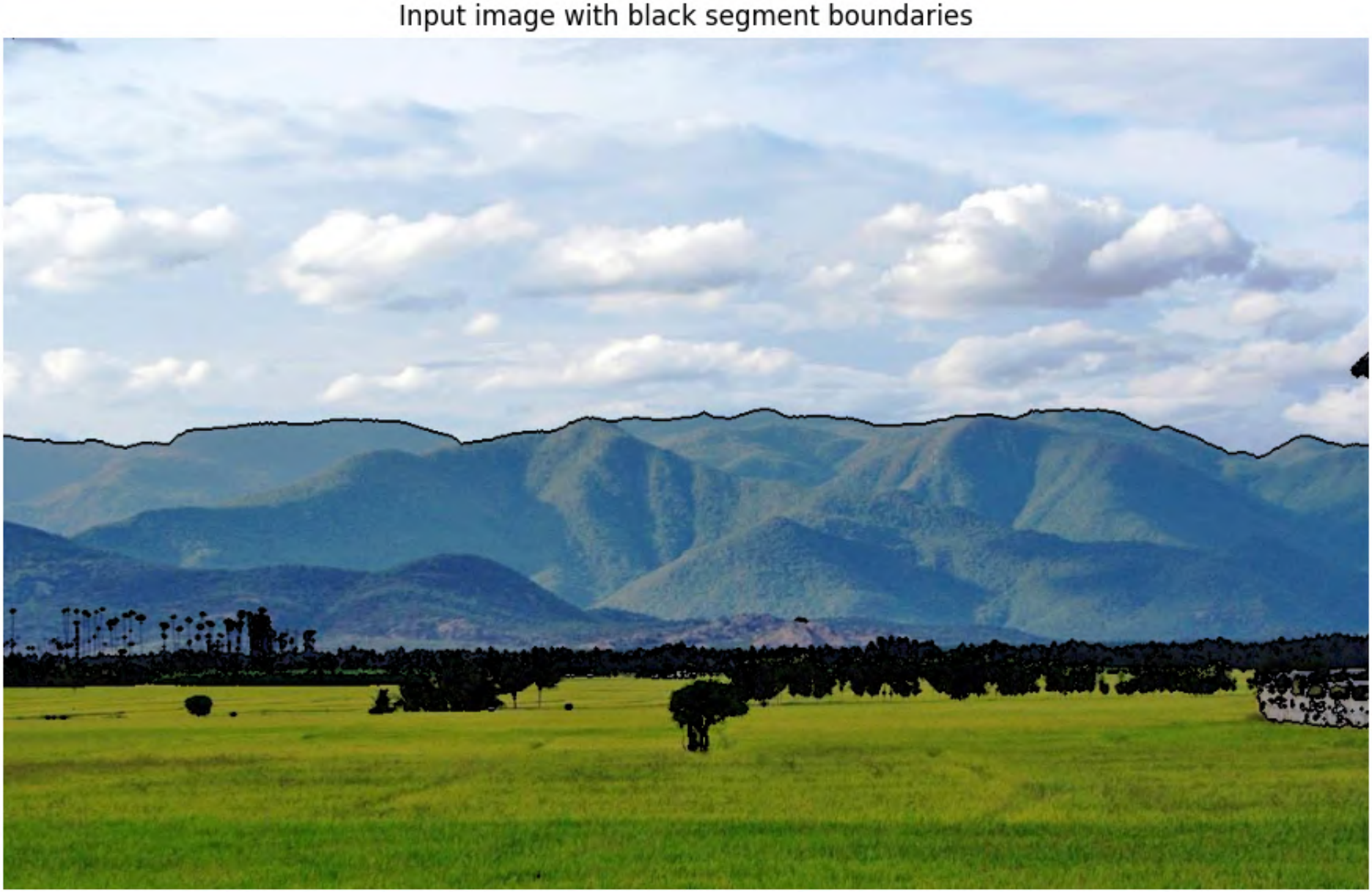
Mean-shift segmentation (4 segments)



Q1 – Ghats: Label Map



Q1 – Ghats: Boundary Overlay



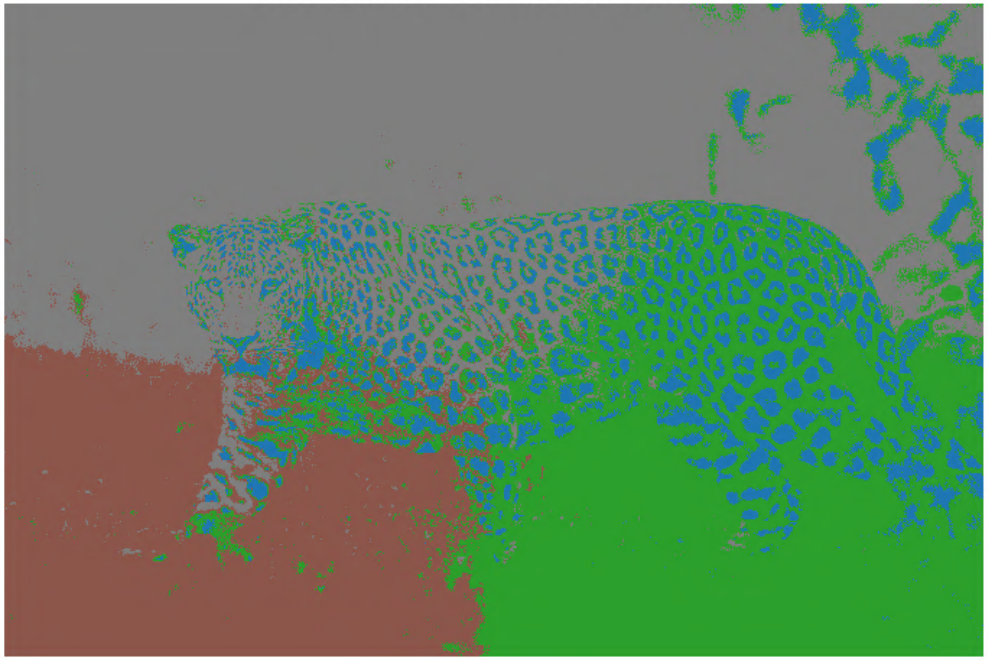
Q1 – Leopard: Segmentation

Mean-shift segmentation (hr=25, hs=300)

Input image



Mean-shift segmentation (4 segments)



Q1 – Leopard: Label Map



Q1 – Leopard: Boundary Overlay

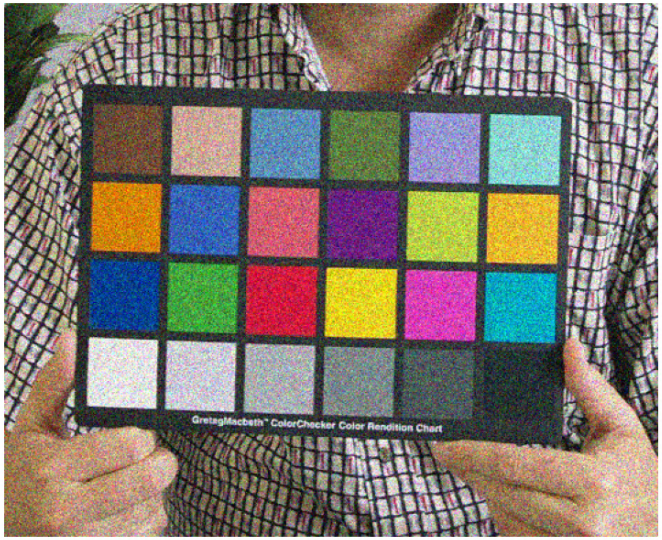
Input image with black segment boundaries



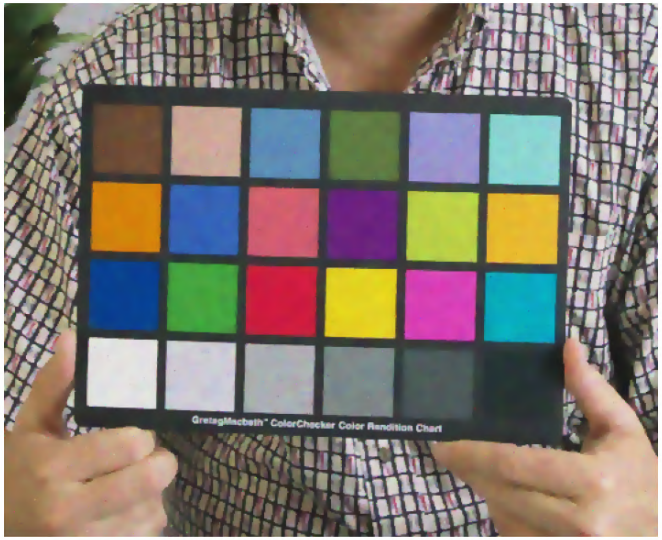
Q2 – Color Checker: Original, Noisy and Denoised Comparison

colorChecker

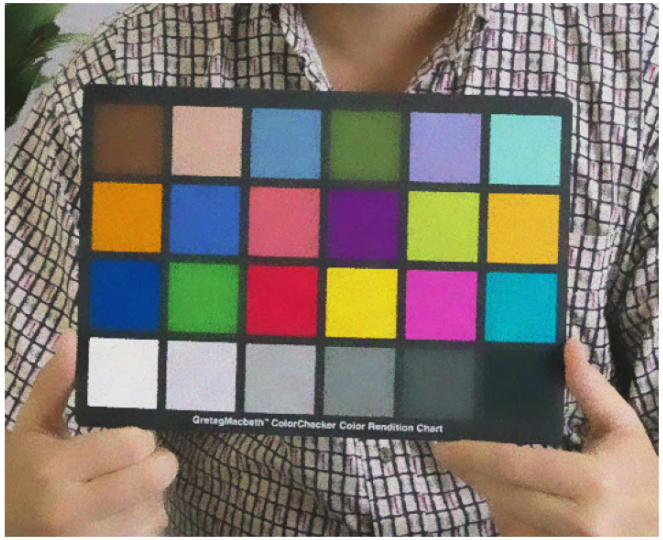
Input image



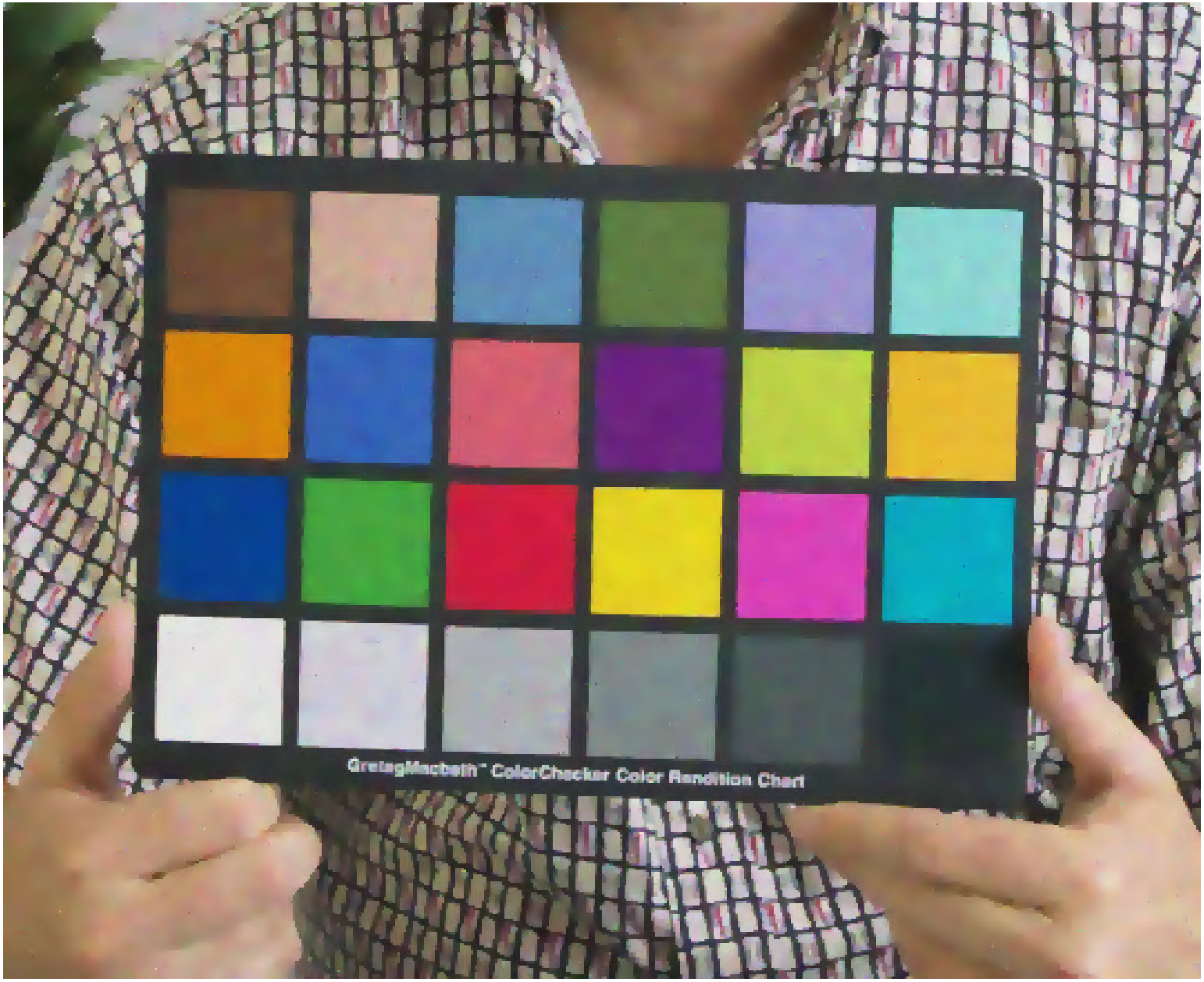
Anisotropic diffusion



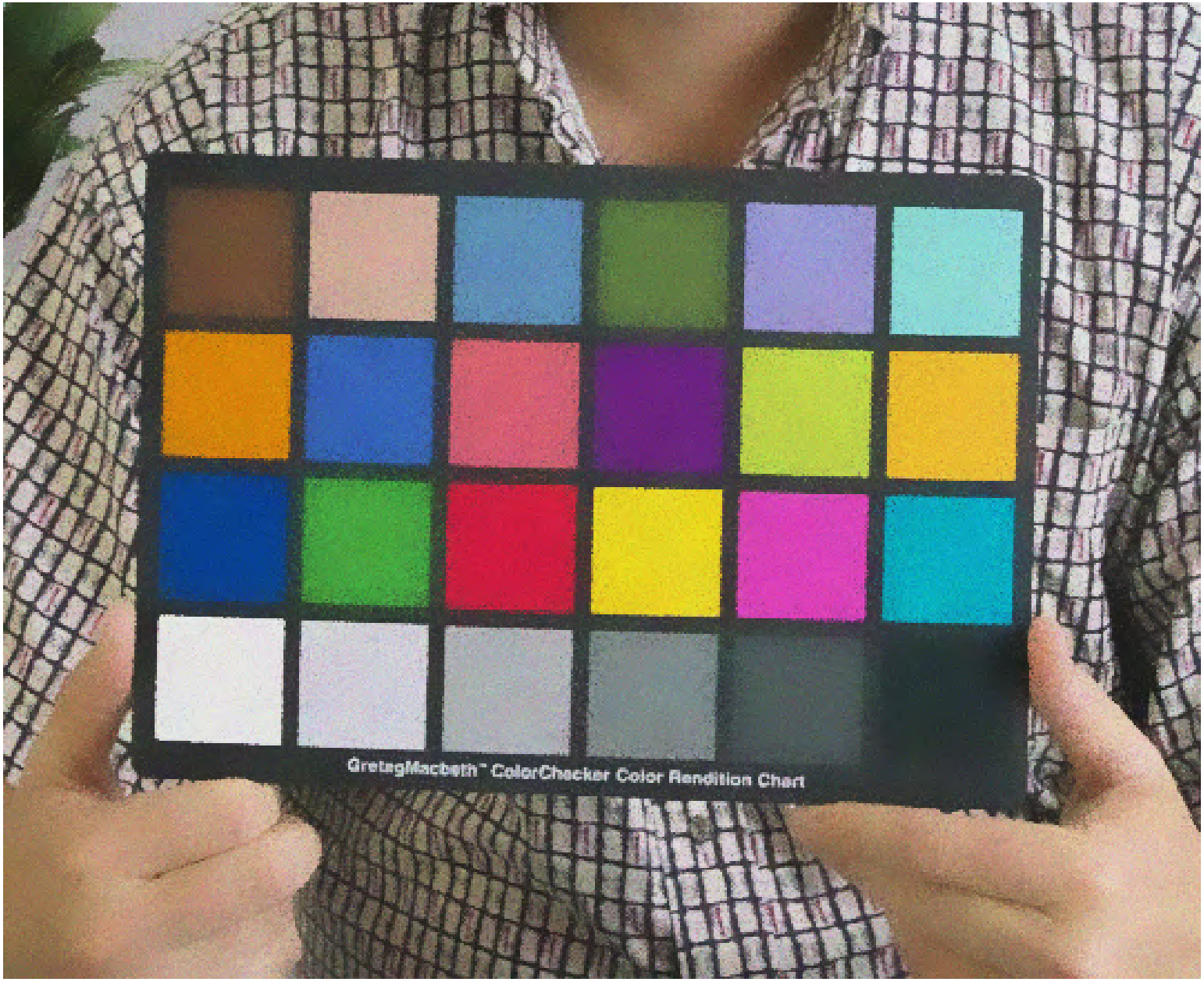
Bilateral filtering



Q2 – Color Checker: Anisotropic Diffusion

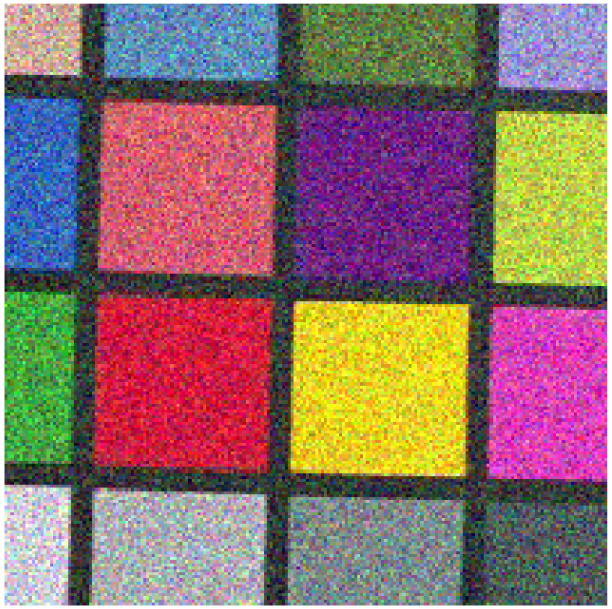


Q2 – Color Checker: Bilateral Filtering



Q2 – Color Checker: Zoomed Comparison

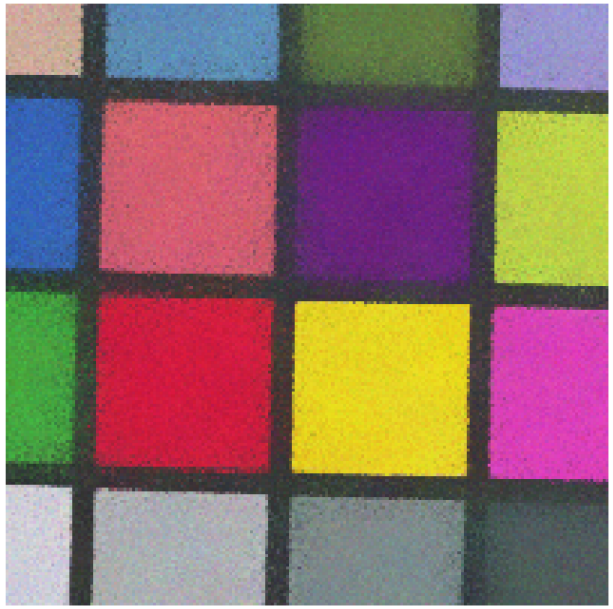
Input image (zoom)



colorChecker: enlarged central crop
Anisotropic diffusion (zoom)



Bilateral filtering (zoom)



Q2 – Raja Ravi Varma: Original, Noisy and Denoised Comparison

rajaRaviVarma

Input image



Anisotropic diffusion



Bilateral filtering



Q2 – Raja Ravi Varma: Anisotropic Diffusion



Q2 – Raja Ravi Varma: Bilateral Filtering



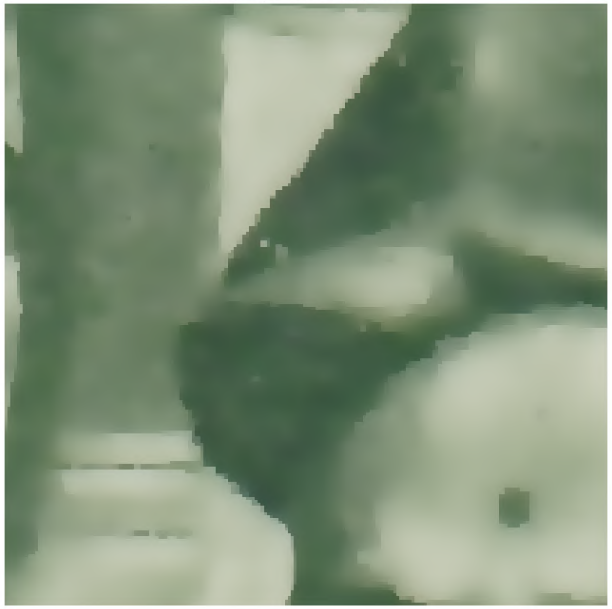
Q2 – Raja Ravi Varma: Zoomed Comparison

rajaRaviVarma: enlarged central crop

Input image (zoom)



Anisotropic diffusion (zoom)



Bilateral filtering (zoom)



Q2 – Sagarmatha: Original, Noisy and Denoised Comparison

sagarmaathaa



Q2 – Sagarmatha: Anisotropic Diffusion



Q2 – Sagarmatha: Bilateral Filtering



Q2 – Sagarmatha: Zoomed Comparison

Input image (zoom)



sagarmaathaa: enlarged central crop
Anisotropic diffusion (zoom)



Bilateral filtering (zoom)

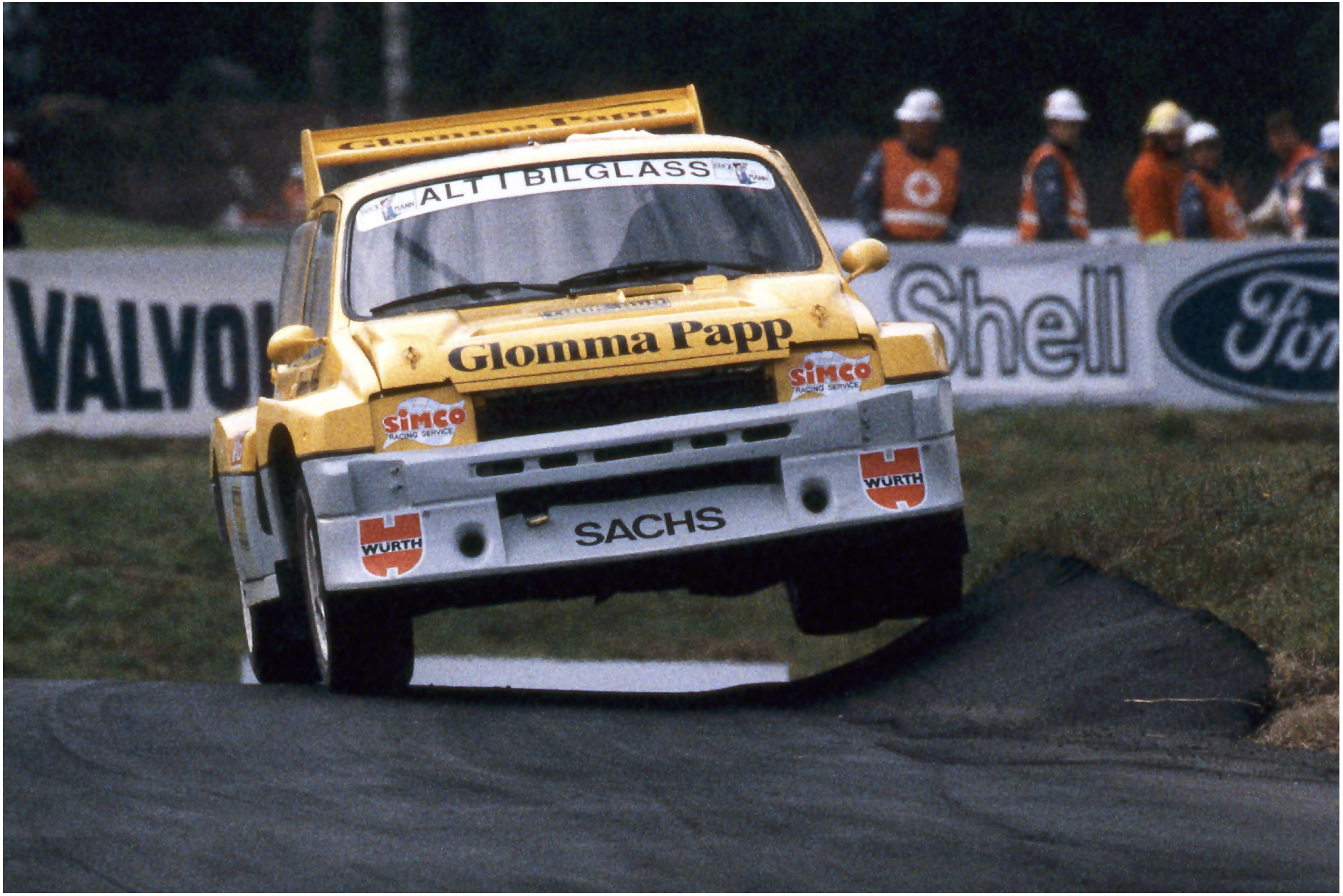


Q2 – Racing: Original, Noisy and Denoised Comparison

racing



Q2 – Racing: Anisotropic Diffusion



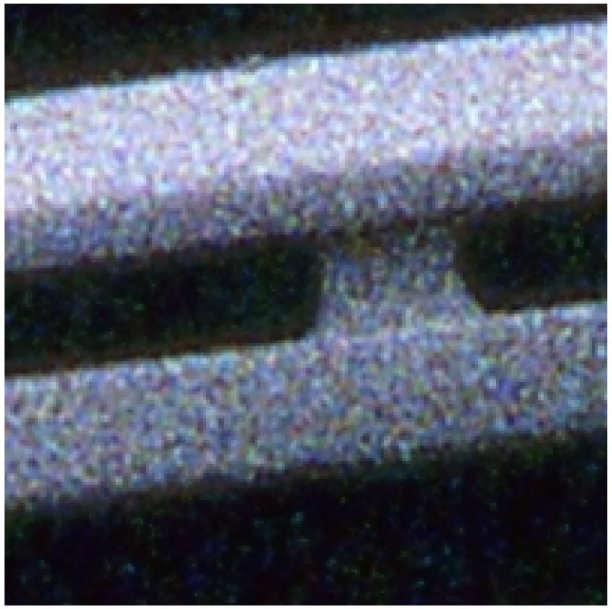
Q2 – Racing: Bilateral Filtering



Q2 – Racing: Zoomed Comparison

racing: enlarged central crop

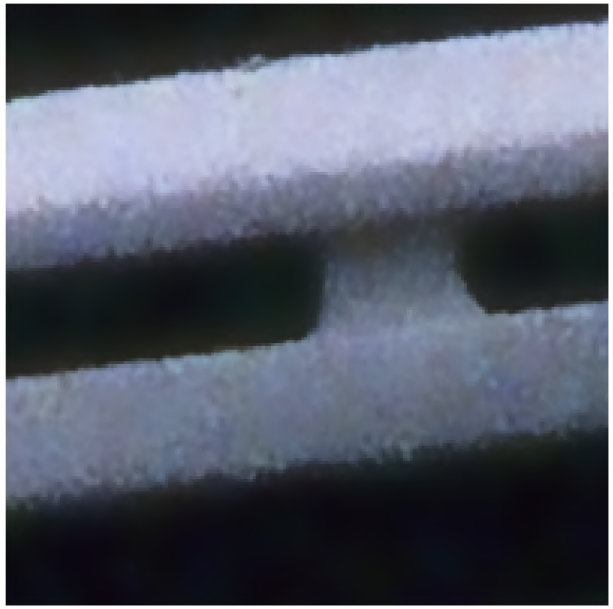
Input image (zoom)



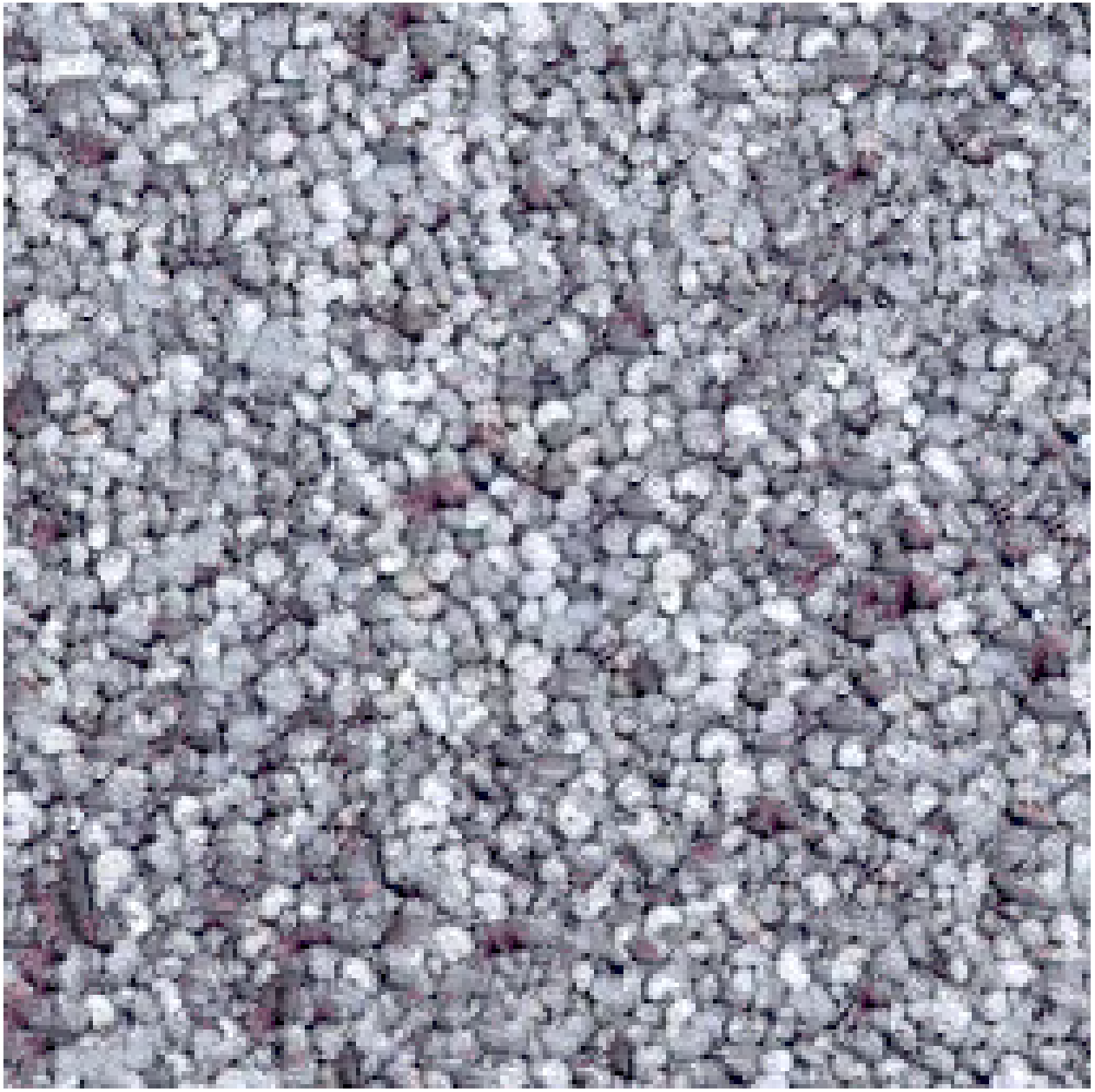
Anisotropic diffusion (zoom)



Bilateral filtering (zoom)



Q3 – Texture Sample 81: Original and Synthesized Texture



Q3 – Texture Sample 110: Original and Synthesized Texture



Q3 – Texture Sample 207: Original and Synthesized Texture



Q3 – Lohagad: Original Image with Hole and Inpainted Result

Original Image



Generated Mask

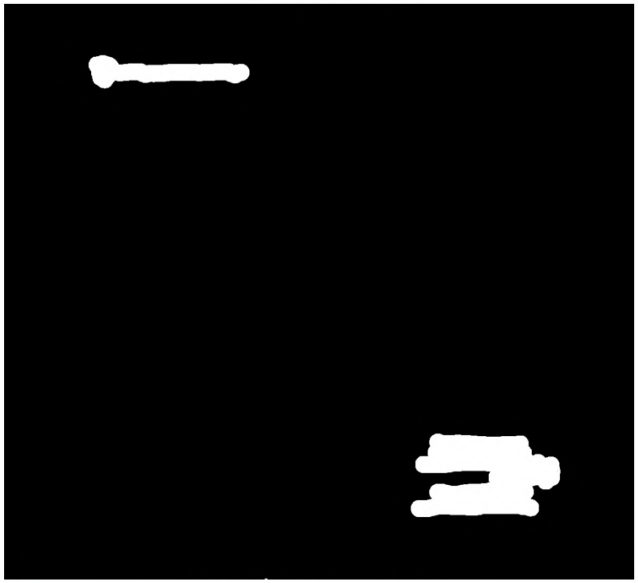


Image with Hole



Texture-Based Inpainting



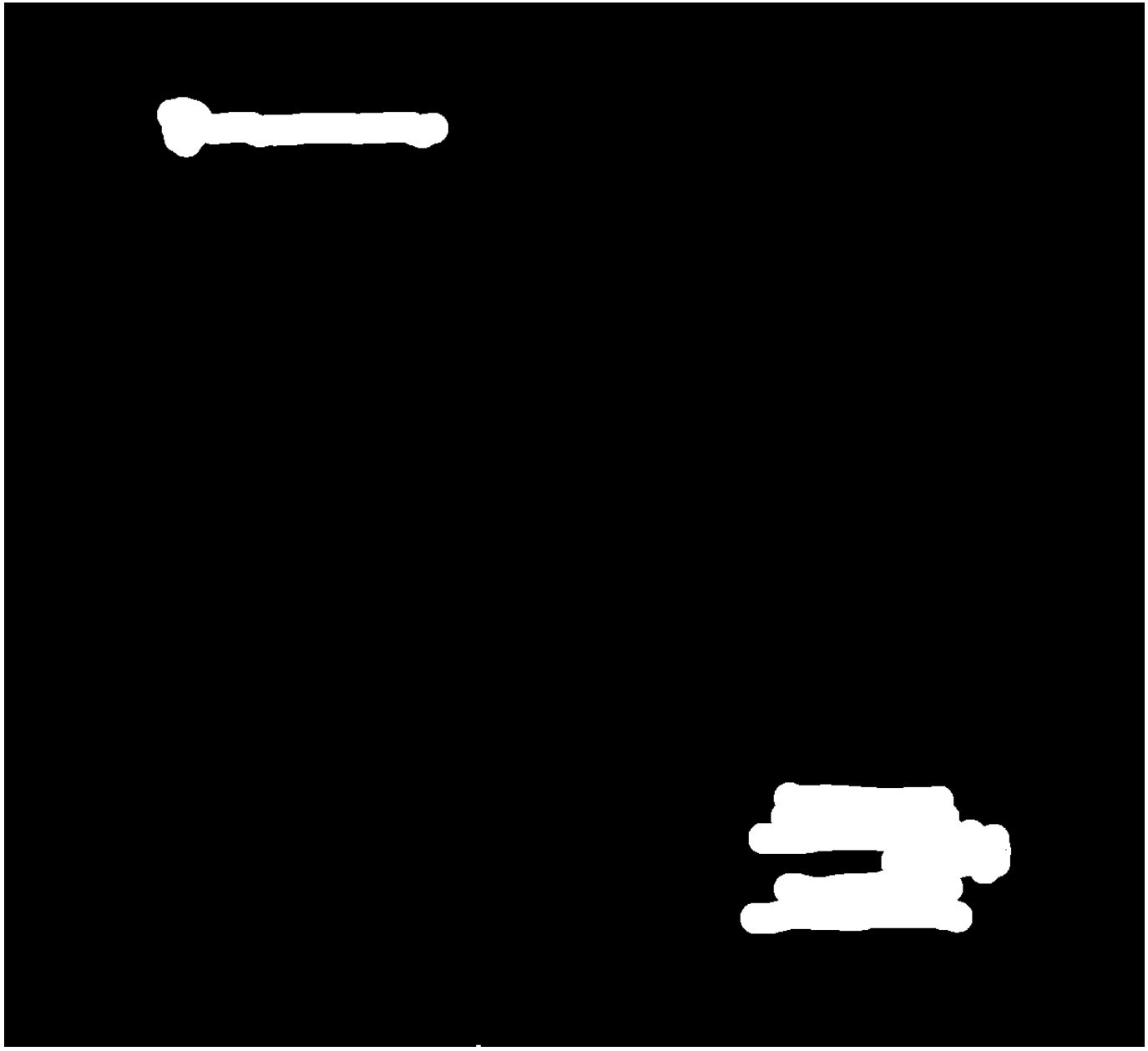
Q3 – Lohagad: Image with Missing Region (Hole)



Q3 – Lohagad: Inpainted Image

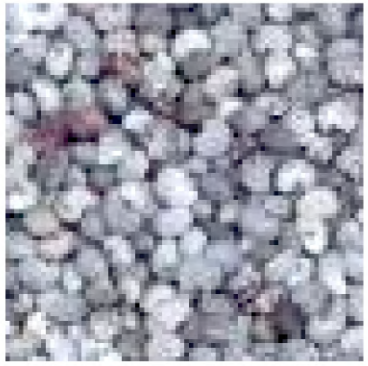


Q3 – Lohagad: Text Removal Mask



Q3 – Texture Synthesis: Comparison of Synthesized Results

Sample: textureSample81.png



Synthesized (256x256)



Sample: textureSample110.png



Synthesized (256x256)



Sample: textureSample207.png



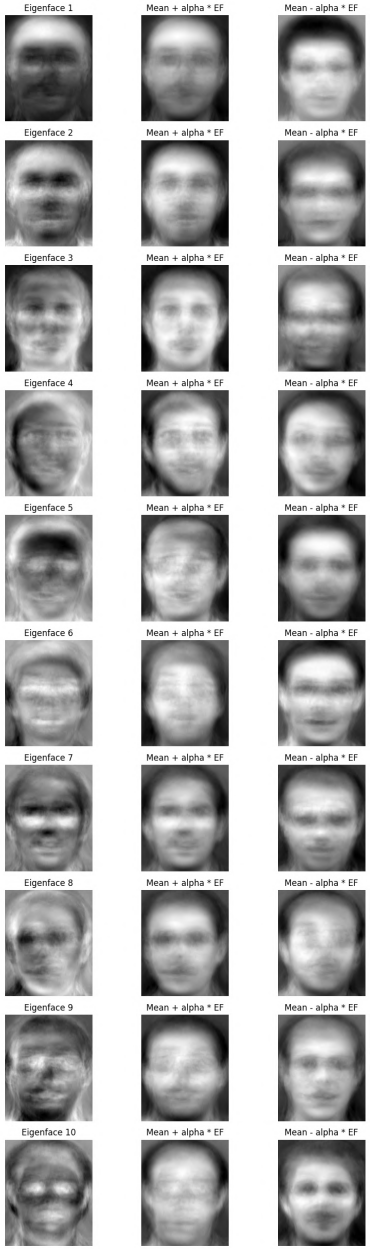
Synthesized (256x256)



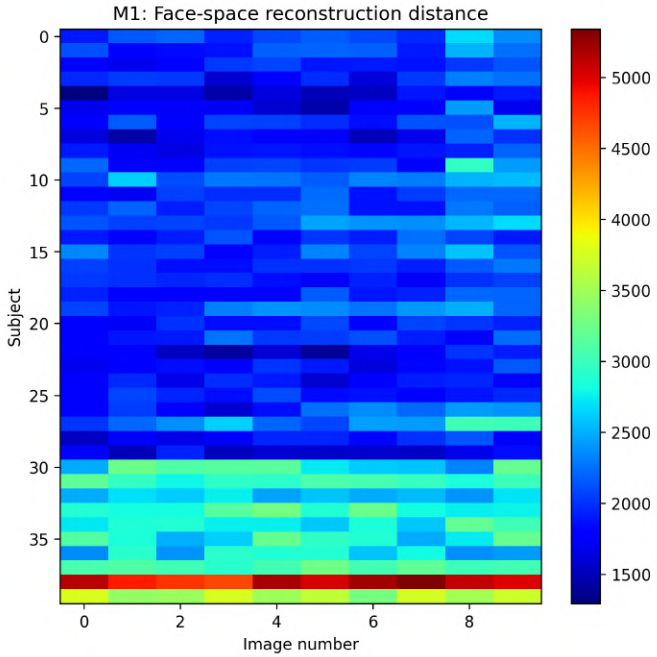
Q4 – Mean Face Image

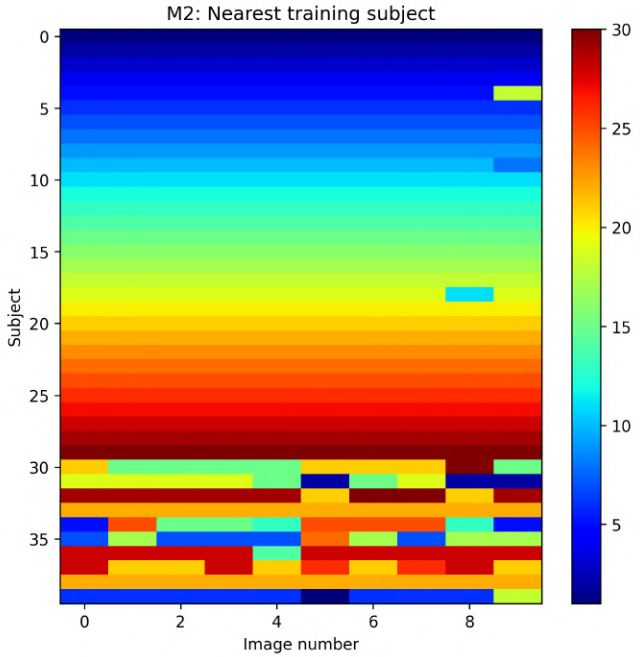


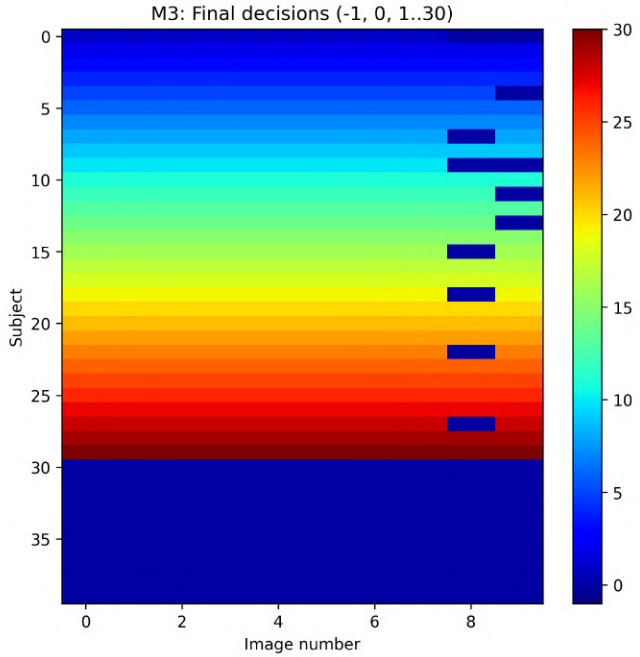
Q4 – First 10 Principal Modes of Variation



Q4 – Eigenface and Mean Image Visualization







Q4 – M1, M2 and M3: Matrix Visualizations

